	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 4 – Organic Chemistry and Spectroscopic Analysis

Date: Tuesday July 13, 2019

Reading Time	5 minutes
Writing Time	55 minutes

Task Sections	Marks	Your Marks
Total Marks	56	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

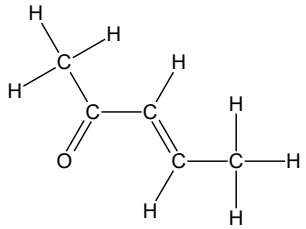
- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.
- You may keep the data book.

Electronic Devices

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Question 1 [2 marks]

Carefully draw a full structure, ***showing all bonds***, that, with respect to the structure at the right, is

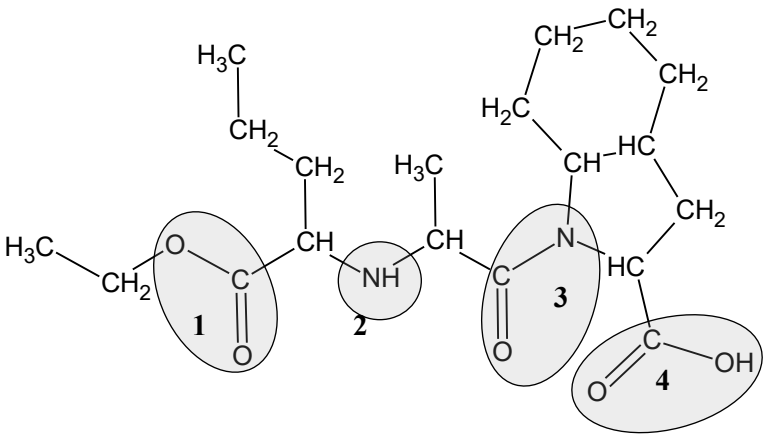


a) a structural isomer (1 mark)

b) a stereoisomer (geometrical isomer) (1 mark)

Question 2 [6 marks]

Below is the structure of perindopril, a drug frequently prescribed to treat high blood-pressure.



a) ***Carefully*** label ***ALL*** of the *chiral centres* in this molecule with an asterisk (*). (2 marks)

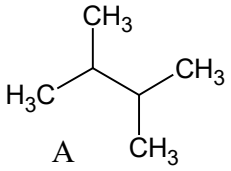
b) This molecule contains four of the functional groups we have studied. Identify each of the four different functional groups indicated. (4 marks)

1 _____ 2 _____

3 _____ 4 _____

Question 3 [11 marks]

Consider the following organic structures: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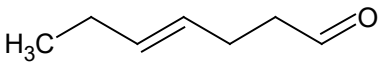
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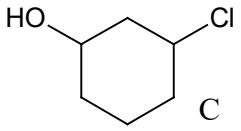
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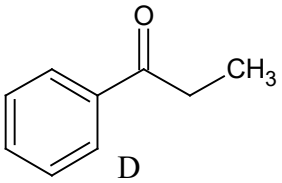
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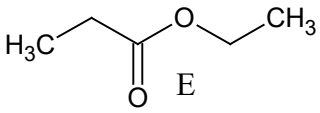
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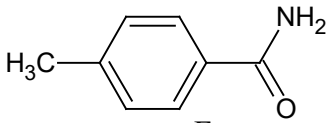
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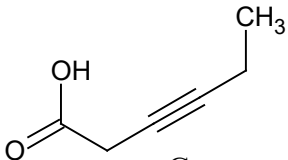
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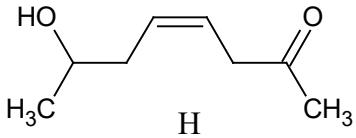
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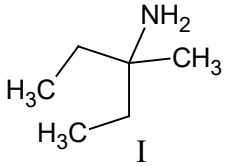
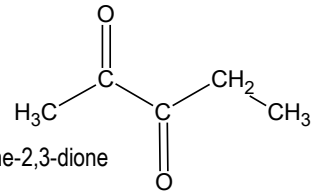

I

In each case below, circle the letters for ***ALL of the structures*** that match the description, or circle NONE if there is no structure that matches. Some structures/letters will be circled more than once. (11 marks)

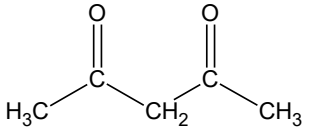
a) an ester	A	B	C	D	E	F	G	H	I	NONE
b) a trans- geometrical isomer	A	B	C	D	E	F	G	H	I	NONE
c) an aromatic compound	A	B	C	D	E	F	G	H	I	NONE
d) an alkyne	A	B	C	D	E	F	G	H	I	NONE
e) an alcohol (hydroxyl group)	A	B	C	D	E	F	G	H	I	NONE
f) a ketone	A	B	C	D	E	F	G	H	I	NONE
g) has at least one chiral centre	A	B	C	D	E	F	G	H	I	NONE
h) a carboxylic acid	A	B	C	D	E	F	G	H	I	NONE
i) an amine	A	B	C	D	E	F	G	H	I	NONE
j) a substituted cyclohexane	A	B	C	D	E	F	G	H	I	NONE
k) a structural isomer of hexane	A	B	C	D	E	F	G	H	I	NONE

Question 4 [6 marks]

The chemical storeroom stocks the two isomers of pentanedione (di-ketones) shown below:



pentane-2,3-dione



pentane-2,4-dione

Unfortunately the labels have faded and because the two bottles were stored next to each other they have become indistinguishable. To identify them ^{1}H and ^{13}C NMR spectra were recorded for samples from each bottle.

Explain how each of the following features of these spectra would have been useful in distinguishing the two compounds. In each case justify your answer, including all relevant details the spectra would contain.

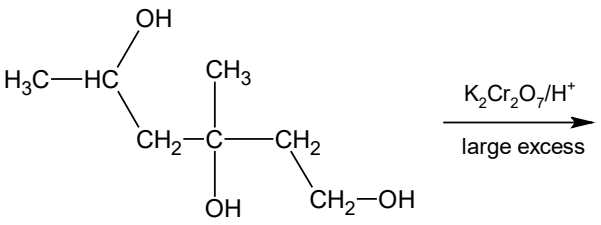
a) The ***number*** of signals (ignoring chemical shift data) in the ^{13}C NMR spectrum of each compound. (2 marks)

b) The ***number*** of low resolution signals (that is, ignoring any peak splitting and any chemical shift data) in the ^{1}H NMR spectrum of each compound, and the ***peak integrals*** (the number of H atoms contributing to each peak). (2 marks)

c) The ***splitting patterns*** observed in the high resolution ^{1}H NMR spectrum of each compound. (2 marks)

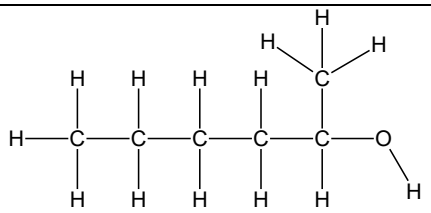
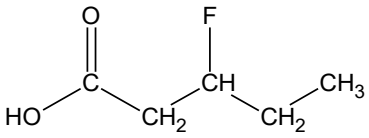
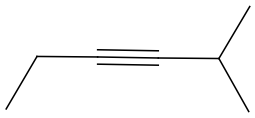
Question 5 [2 marks]

Draw the structure of the likely organic product formed when the compound shown below is heated with a large excess of acidified potassium dichromate until no further reaction is observed. (2 marks)



Question 6 [4 marks]

Give a systematic name for each of the structures shown below. (4 marks)

	
	
	
$CH_3(CH_2)_3COO(CH_2)_2CH_3$	

Question 7 [4 marks]

For each compound named below draw both a complete structure, ***showing all atoms and bonds***, and also a skeletal structure.

a) *cis*-5-methylhex-2-ene (2 marks)

full structure	skeletal structure

b) 3-chlorobutan-2-amine (2 marks)

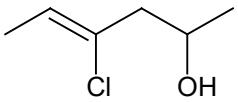
full structure	skeletal structure

Question 8 [2 marks]

Write a semi-structural formula for each of the following. (2 marks)

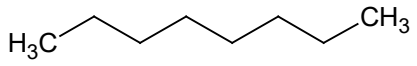
a) 5-aminoheptanoic acid

b)



Question 9 [3 marks]

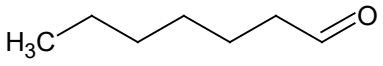
Carefully account for the different boiling points observed for the three compounds below. (3 marks)



octane

M = 114 g mol^{-1}

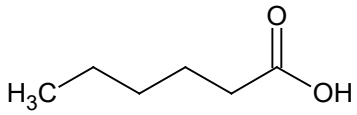
bp = 126 °C



heptanal

M = 114 g mol^{-1}

bp = 153 °C



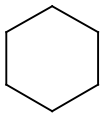
hexanoic acid

M = 116 g mol^{-1}

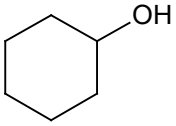
bp = 205 °C

Question 10 [3 marks]

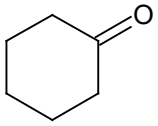
Consider the following four structures.



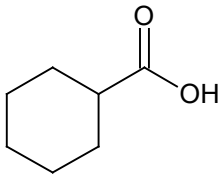
A



B

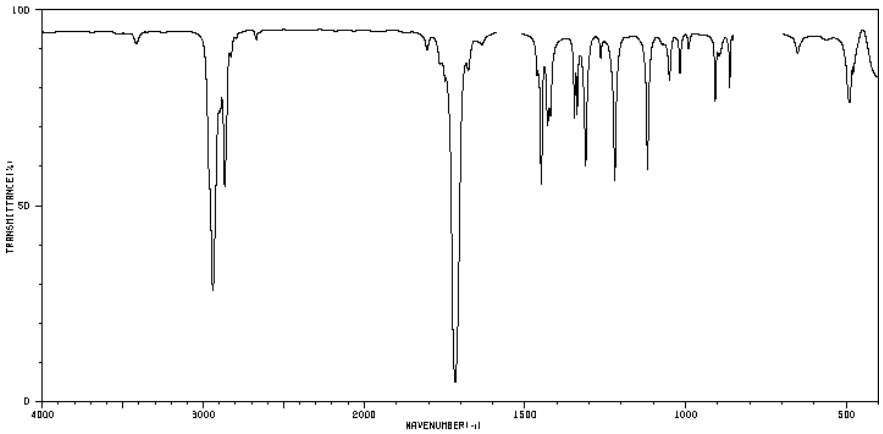


C



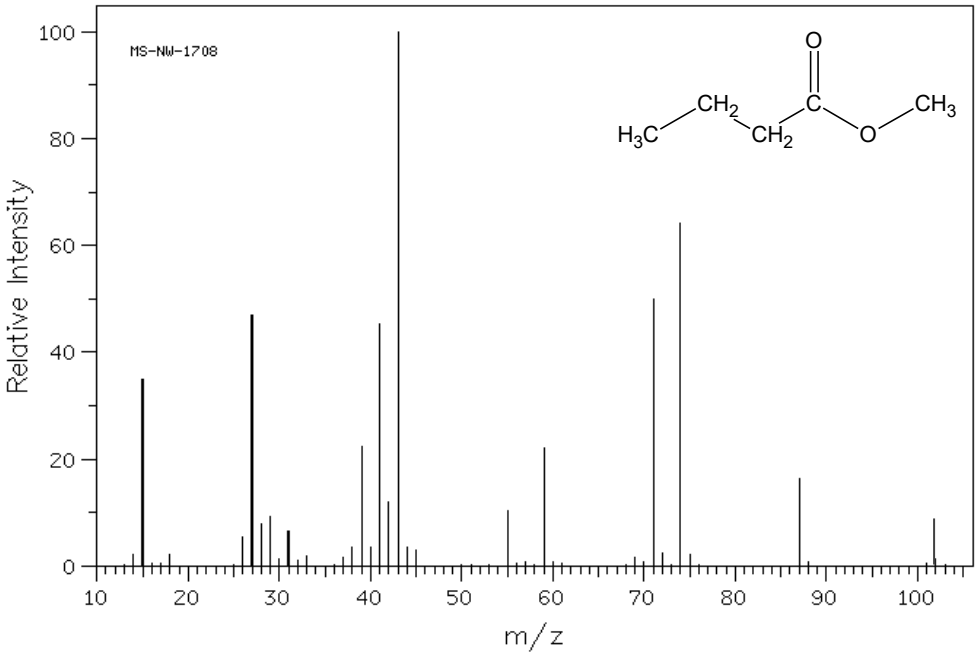
D

Which of these is most likely to have produced the following IR spectrum? Carefully justify your choice, by referring to all relevant conclusions that can be drawn from the spectrum. (3 marks)



Question 11 [4 marks]
At the right is a mass spectrum of methyl butanoate.

Refer to the spectrum and answer the following questions. (4 marks)



a) What are the m/z values for
   i. the parent/molecular ion peak? .....

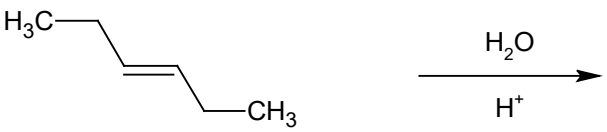
   ii. the base peak? .....

b) The signals in the spectrum that are listed below can be attributed to fragments arising from simple fission (breaking) of one bond in the parent ion. Write the formula of the particle likely to be responsible for the signal at
   i. m/z = 43 .....

   ii. m/z = 71 .....

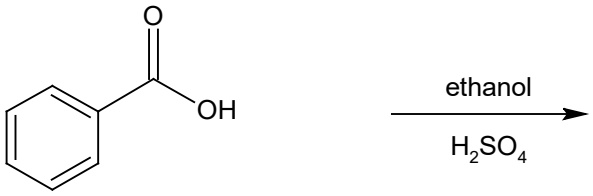
Question 12 [4 marks]
Draw the main organic products expected from the following reactions, and in each case state the type of reaction occurring. (2+2 marks)

a)



REACTION TYPE: .....

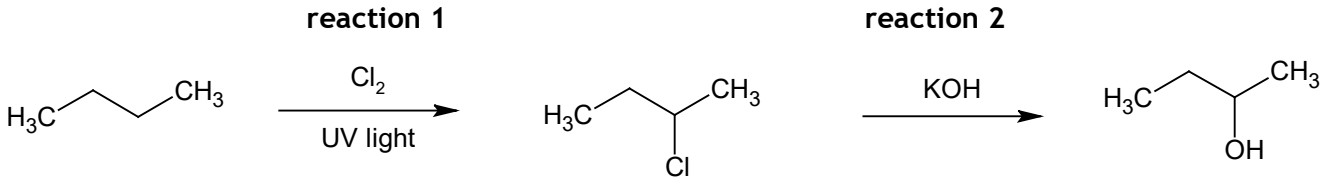
b)



REACTION TYPE: .....

Question 13 [5 marks]

Consider the following reaction sequence for the preparation of butan-2-ol from butane.



a) What type of reaction is the chlorination depicted in reaction 1? (1 mark)

b) In one case, starting with 350 g of butane, 320 g of butan-2-ol was obtained. What was the percentage yield? (2 marks)

c) The overall process can be represented by the following balanced equation:

$$C_4H_{10} + Cl_2 + 2KOH \rightarrow C_4H_9OH + 2KCl + H_2O$$

Calculate the atom economy of this process. (2 marks)

End of Task – 56 marks available